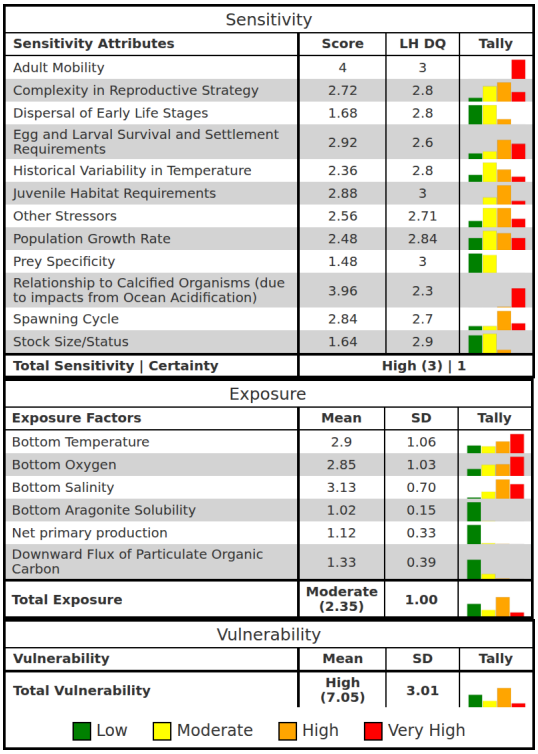
Atlantic surfclam (*Spisula solidissima*)

Summary Scores

- **Sensitivity:** High
- **Exposure:** Moderate
- **Vulnerability:** High



Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	4	3	
Complexity in Reproductive Strategy	2.72	2.8	
Dispersal of Early Life Stages	1.68	2.8	
Egg and Larval Survival and Settlement Requirements	2.92	2.6	
Historical Variability in Temperature	2.36	2.8	
Juvenile Habitat Requirements	2.88	3	
Other Stressors	2.56	2.71	
Population Growth Rate	2.48	2.84	
Prey Specificity	1.48	3	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	3.96	2.3	
Spawning Cycle	2.84	2.7	
Stock Size/Status	1.64	2.9	
Total Sensitivity | Certainty	**High (3) | 1**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	2.9	1.06	
Bottom Oxygen	2.85	1.03	
Bottom Salinity	3.13	0.70	
Bottom Aragonite Solubility	1.02	0.15	
Net primary production	1.12	0.33	
Downward Flux of Particulate Organic Carbon	1.33	0.39	
Total Exposure	**Moderate (2.35)**	**1.00**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**High (7.05)**	**3.01**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Atlantic surfclam. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 6 and Figure 5). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	High	High	None
Exposure	High	Moderate	Decrease
Vulnerability	High	High	None
Directional Effect	Negative	Negative	None

Stocks

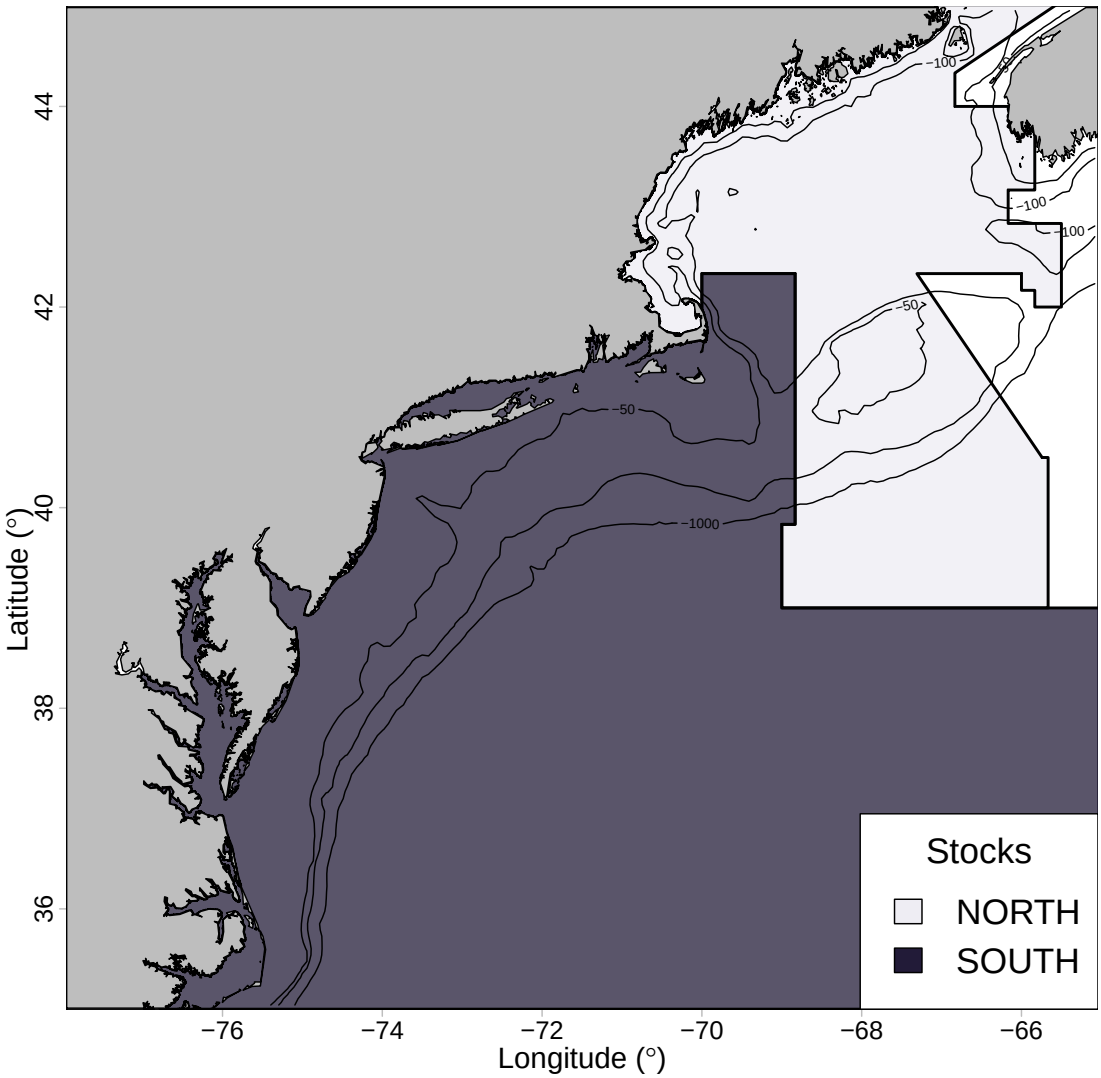


Figure 2: Atlantic surfclam Stocks along the Northeast Shelf

Score Descriptions:

Sensitivity:

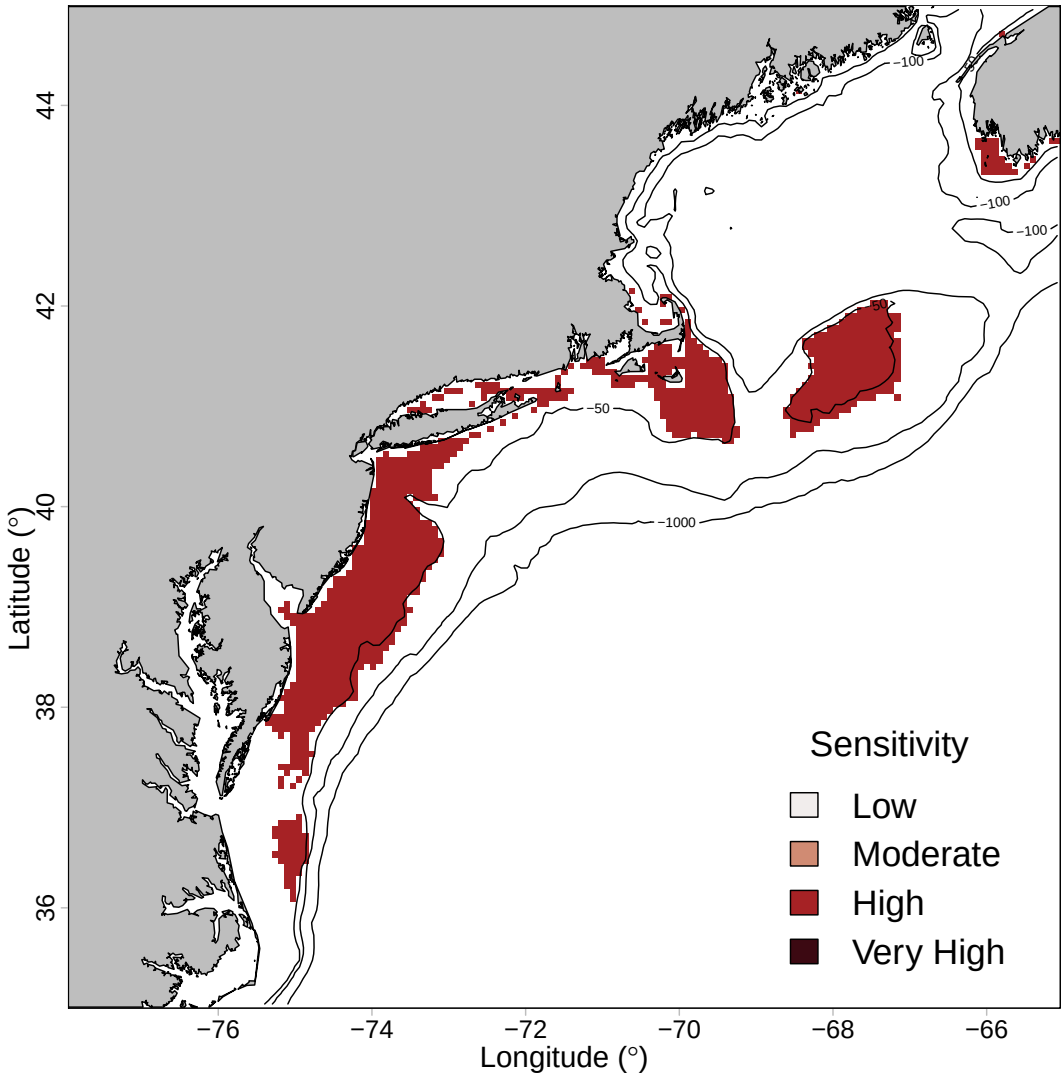


Figure 3: Atlantic surfclam Sensitivity on the Northeast Shelf

Atlantic surfclams received a **High** sensitivity score, with 100% certainty (Figure 1). Two attributes had scores greater than 3: Adult Mobility (4) and Relationship to Calcified Organisms (4) (Figure 1). Like other bivalves, they have a calcium carbonate shell and are therefore susceptible to ocean acidification (Cargnelli et al. 1999). In addition, they are primarily sessile, other than burrowing into sediments to avoid predators (Cargnelli et al. 1999).

Surfclams received the same sensitivity score in the 2016 Climate Vulnerability Assessment. Adult Mobility and Relationship to Calcified Organisms also drove that score (Hare et al. 2016).

Exposure:

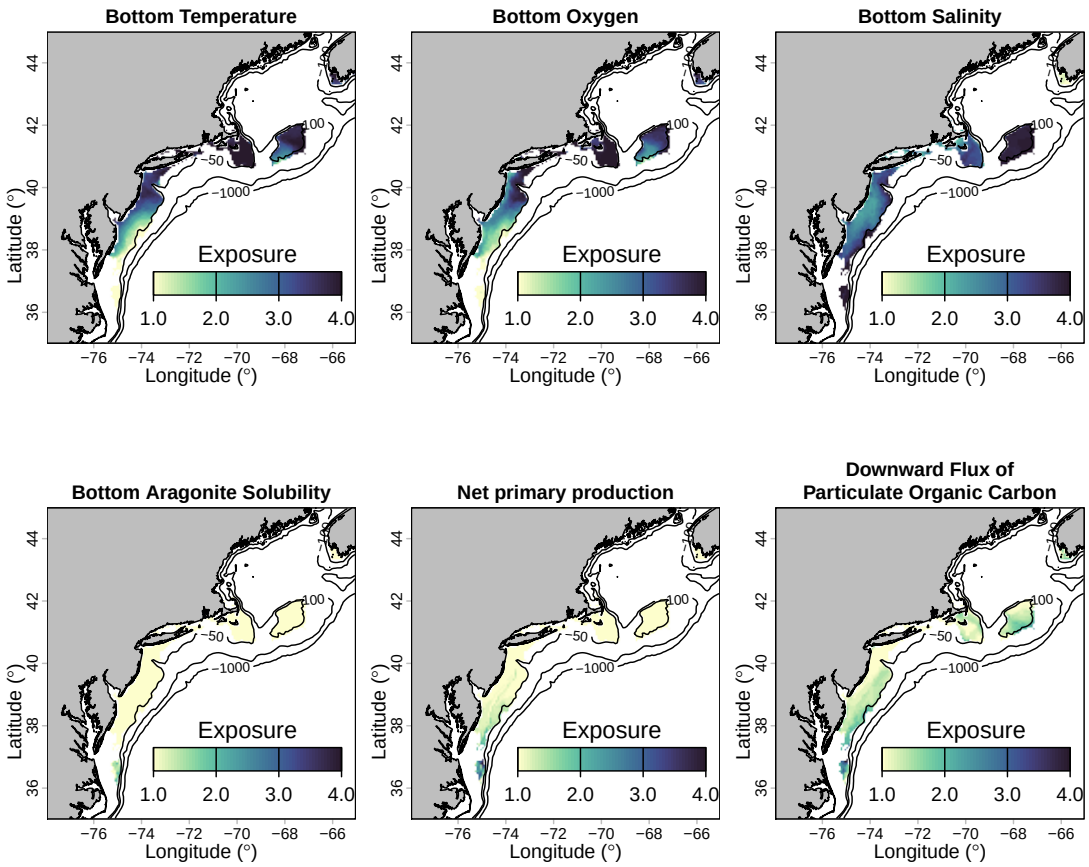


Figure 4: Maps of each Exposure Factor for Atlantic surfclam. Exposure was calculated between 2014-2023 and 2025-2035

The Atlantic surfclam had a mean Exposure score of 2.35 +/- 1.0, resulting in a **Moderate** average score throughout the Northeast Shelf (Figure 1). This was driven by high exposure to bottom temperature, oxygen, and salinity changes (Figure 4). Exposure was highest on Georges Bank and generally decreased moving down the coastline (Figure 5).

In the previous Assessment, surfclam received a High exposure score, driven by sea surface temperature and ocean acidification (Hare et al. 2016).

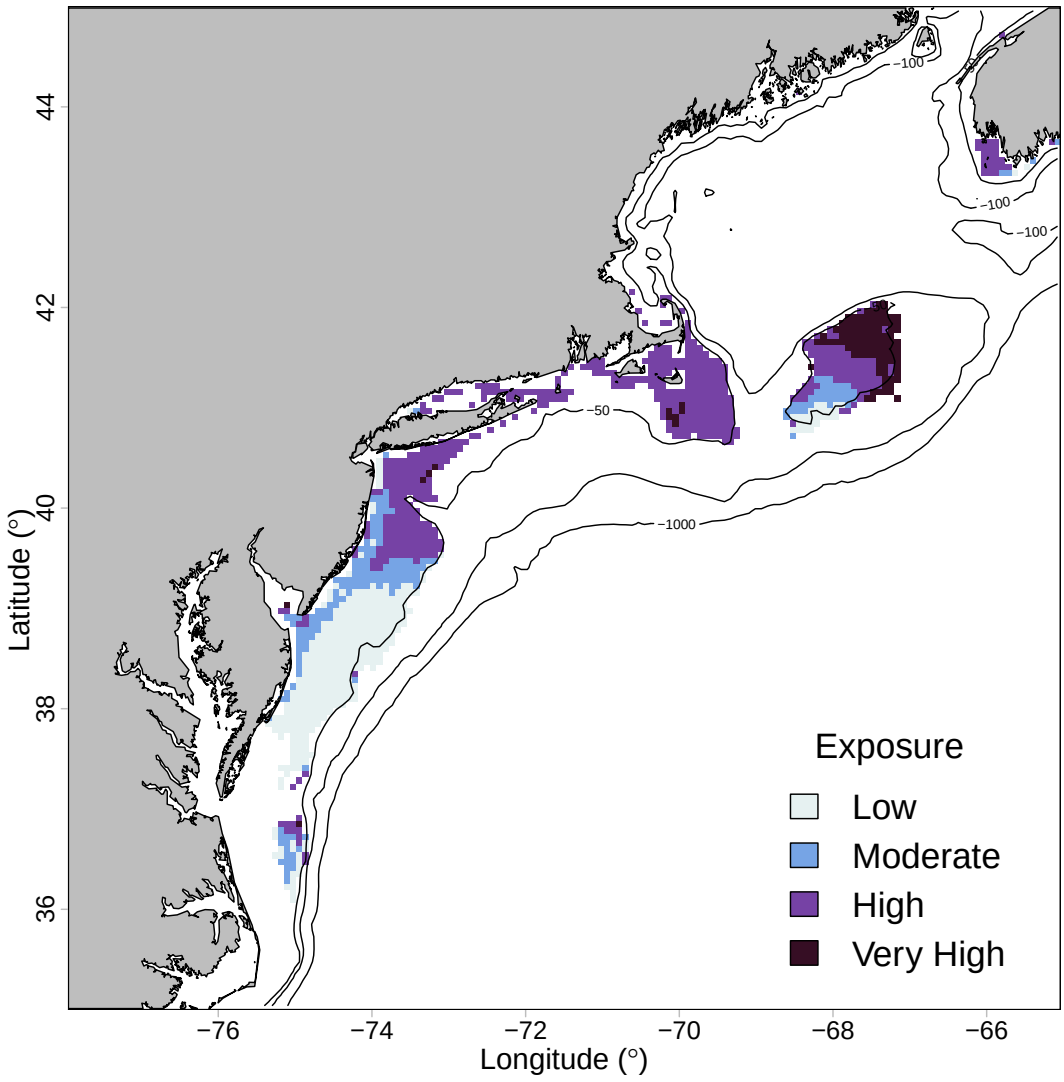


Figure 5: Total Exposure on the Northeast Shelf for Atlantic surfclam. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 4)

Vulnerability:

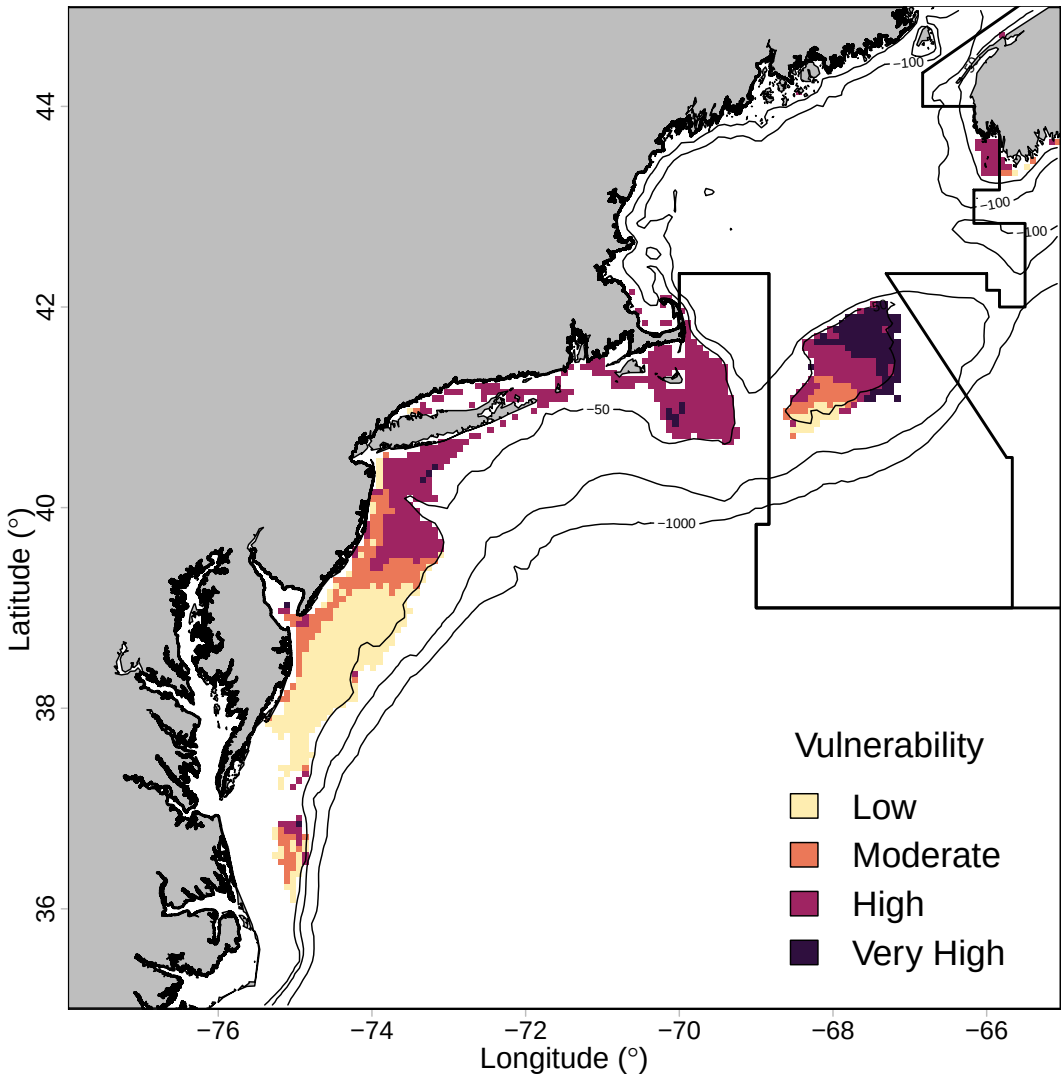


Figure 6: Vulnerability on the Northeast Shelf for Atlantic surfclam. Vulnerability is the product of sensitivity and exposure (see Figure 5 and Figure 3)

The mean overall Vulnerability rank for Atlantic surfclams was **High** (7.05 +/- 3.01) (Figure 1). Similar to the Exposure score, Vulnerability was highest on Georges Bank and generally decreased moving southward along the coast (Figure 6).

In the previous assessment, Atlantic Surfclam also received a High vulnerability score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

95% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified (Figure 1).

Model Confidence Score:

Mean model confidence for the Atlantic surfclam model was 2.3 (+/- 0.6) indicating moderate confidence in the distribution model. While the model aligned with experts' expectations on the distribution, there were concerns about low sample size. Experts also thought that sediment type should have been included in the models. The ensemble model AUC score was 0.92, indicating excellent predictive performance.

Additional Metrics

Distribution Shifts:

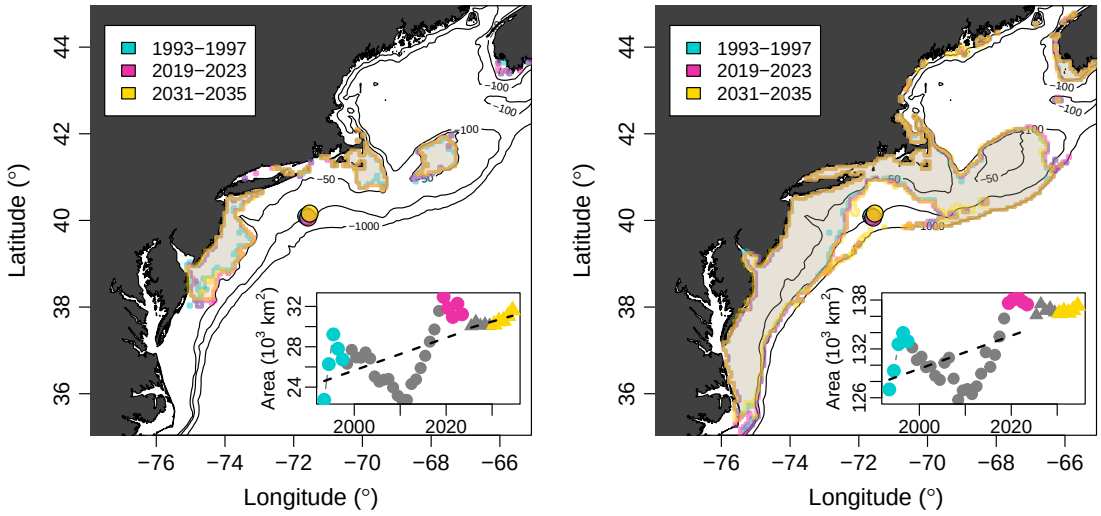


Figure 7: Distribution shifts on the Northeast Shelf for Atlantic surfclam. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Surfclam habitat area experienced a net increase between 1993 and 2023. There was a decline in habitat area in approximately 2010 but the area has since rebounded (Figure 7). Ranges and core habitat expanded further offshore and further south between 1993 and 2023. The weighted centroid shifted approximately 6 km east-southeast from 1993 to 2023.

Core habitat is predicted to continue to grow significantly between 2025 and 2035 due to continued offshore shifts (Figure 7). The distribution centroid is expected to shift 11 km north-northeast.

In the previous Assessment, Atlantic surfclam received a High Distributional Vulnerability Rank (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	26,877	131,735	4.61	8.01
2019-2023	5.7 km ESE	31,880 (+5,004)	137,933 (+6,197)	4.92 (+0.31)	8.10 (+0.08)
2031-2035	11.1 km NNE	30,810 (-1,070)	136,763 (-1,170)	4.42 (-0.50)	8.01 (-0.08)

Directional Effect:

Atlantic surfclams are expected to be **negatively** impacted by climate change. Certainty in expert scores was 80%. Warming bottom waters have shifted surfclam populations from nearshore to more northern and offshore waters (Timbs et al. 2019; Stromp et al. 2023; Hofmann et al. 2018). This northern shift may lead to future competition with ocean quahogs (Spencer et al. 2026; Stromp et al. 2023). Ocean acidification will also likely impact the surfclam. These results are similar to those from the 2016 Climate Vulnerability Assessment. Certainty was higher in the previous assessment (Hare et al. 2016).

Climate Effects on Abundance and Distribution

Warming bottom water temperatures in the Northwest Atlantic have produced a range shift in Atlantic surfclams from southern, inshore waters of the Mid-Atlantic Bight into deeper waters north and offshore (Timbs et al. 2019) (Stromp et al. 2023). The southern range boundary of the species likely began to migrate northward and offshore in the 1970s (Hofmann et al. 2018). A mass mortality event in the inshore waters of Delmarva occurred in 2000, where surfclams were in a "transient event margin", a physiological range where scope for growth is negative but temperatures remain non-lethal, leading to decreased energetics and mortality (Weinberg et al. 2005) (Woodin et al. 2013). Thermal stress in surfclams occurs at 21°C and mortality above 25°C (Munroe et al. 2013) (Powell et al. 2017). Model projections through the end of the 21st century show a positive trend in Atlantic surfclam biomass as their range continues to shift, with biomass expanding in deeper waters off of New Jersey, Long Island, and Southern New England and declines in Georges Bank and Delmarva (Spencer et al. 2026). Warming waters may result in declines in maximum size in the southern portion of the species' range (Powell et al. 2025). This range shift may lead to potential competition between Atlantic surfclams and ocean quahogs (*Arctica islandica*) and changes to the spatial extent of the fishery (Spencer et al. 2026) (Stromp et al. 2023).

Life History Synopsis

Atlantic surfclams are a benthic biomass dominant species on the Northeast U.S. continental shelf from the Gulf of St. Lawrence to Cape Hatteras, North Carolina (Hofmann et al. 2018). The fishery operates in the Mid-Atlantic Bight and Georges Bank and produces over $30 million in exvessel annual revenue (Murray 2016). Spawning occurs in summer to early fall, with earlier spawning occurring in southern habitats (Cargnelli et al. 1999). Temperature is a determinant for gonadal development and spawning (Loosanoff and Davis 1963). Spawning occurs when temperatures exceed 15°C, and may be delayed due to

cold temperatures (Cargnelli et al. 1999). Fertilized eggs develop into planktonic larvae approximately 9 hours after fertilization at 21.7°C and 40 hours at 14°C (Ropes 1980). Larvae mature into the pediveliger stage, a transitional "swimming-crawling" larval stage with development of a foot for burrowing, around 18 days after fertilization (Loosanoff and Davis 1963). Larvae are dispersed by currents (generally southward along the Northeast shelf) and settlement coincides with a relaxation of upwelling events (Fay et al. 1983). Variable reproductive success has been linked to variability in current dynamics in the Mid-Atlantic Bight (Zhang et al. 2016). Metamorphosis to juveniles, which consists of complete absorption of the velum and settlement to the substrate, occurs anywhere from 19 to 35 days after fertilization depending on temperature (Fay et al. 1983). Juveniles settle in sandy habitats in the intertidal and up to depths of 178m (Cargnelli et al. 1999). Juveniles and adults are planktivorous siphon feeders, feeding on phytoplankton 4µm diameter. Food varies with the season, geographic location, and depth of shell bed (Cargnelli et al. 1999). Atlantic surfclams have many predators, including the naticid snails *Euspira heros* and *Neverita duplicata*, the sea star *Asterias forbesi*, lady crabs (*Ovalipes ocellatus*), Jonah crabs (*Cancer borealis*) and horseshoe crabs (*Limulus polyphemus*). Fish predators include haddock (*Melanogrammus aeglefinus*) and Atlantic cod (*Gadus morhua*) (Ropes 1980). Age at maturity ranges anywhere from 3 months to 4 years, however size is likely more important to maturation rates (Cargnelli et al. 1999). Atlantic surfclams have a maximum age of 31 years (Cargnelli et al. 1999). The stock is managed by the Mid-Atlantic Fishery Management Council under the Atlantic surfclam and ocean quahog Fishery Management Plan. As of the 2024 Management Track Assessment, the stock is not overfished and overfishing is not occurring. However, spawning stock biomass continues to decline and is currently at 95% of the target biomass (Northeast Fisheries Science Center (NEFSC) 2024).

Supplemental Scores

Exposure Timeseries

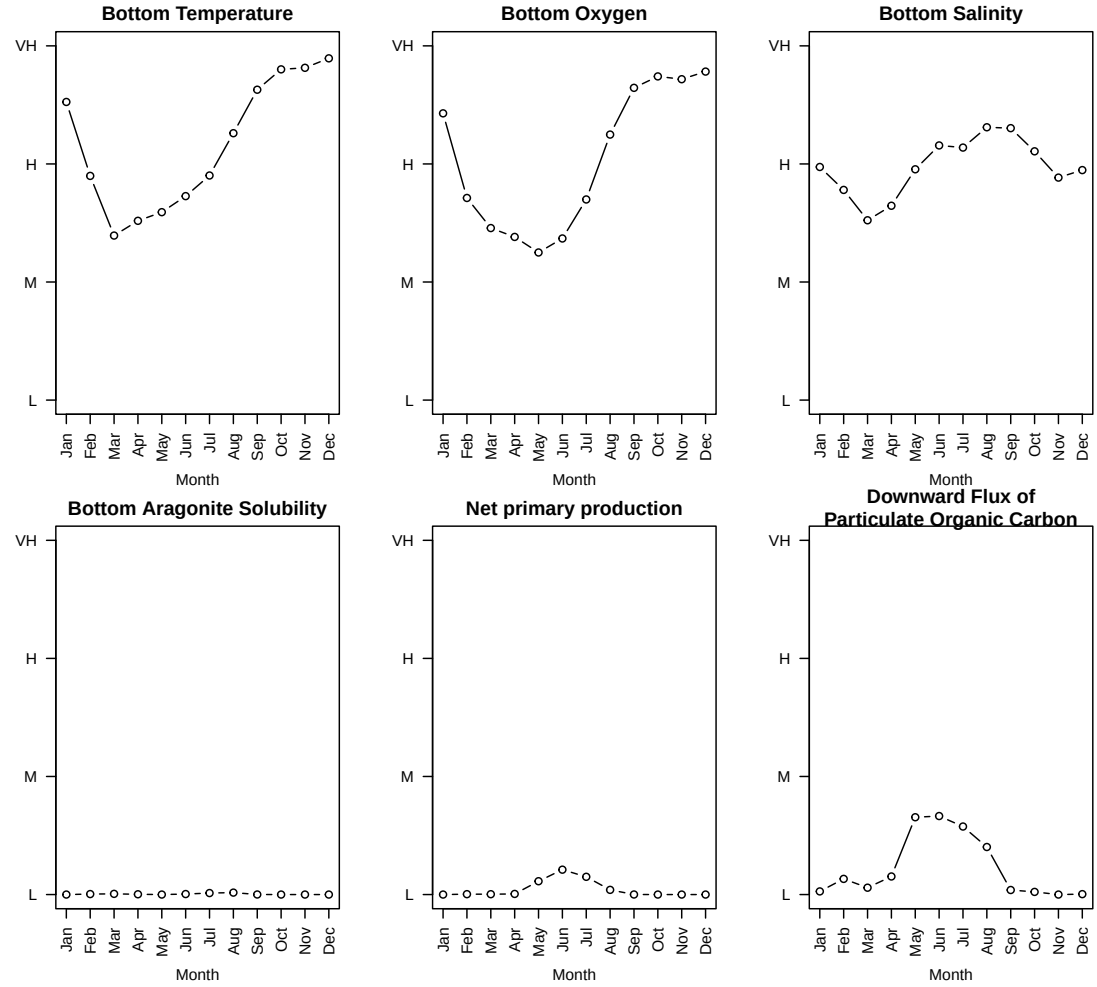


Figure 8: Monthly Timeseries of each Exposure Factor for Atlantic surfclam. Exposure was calculated between 2014-2023 and 2025-2035

Monthly exposure ranged from Low to High. Exposure was Low in March, and High from August through January (Figure 9). These patterns were driven by bottom temperature, oxygen, and salinity exposure (Figure 8).

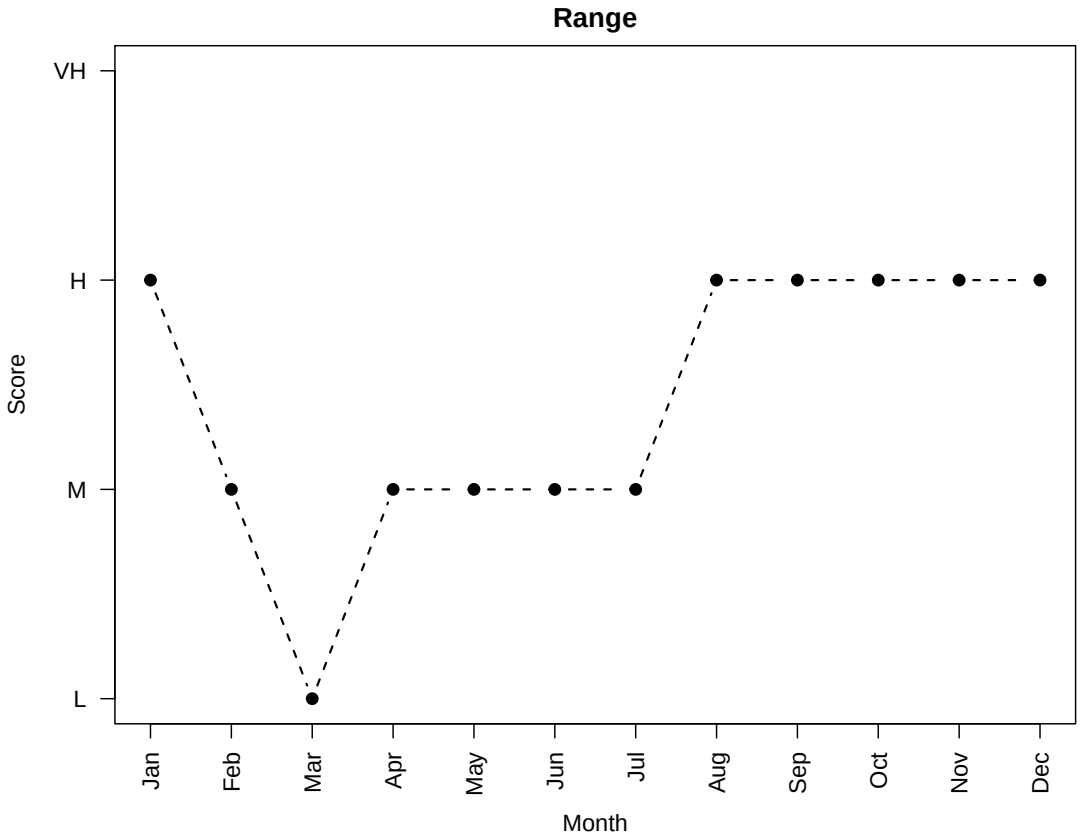


Figure 9: Timeseries of Total Exposure for Atlantic surfclam

Exposure due to Significant Factors

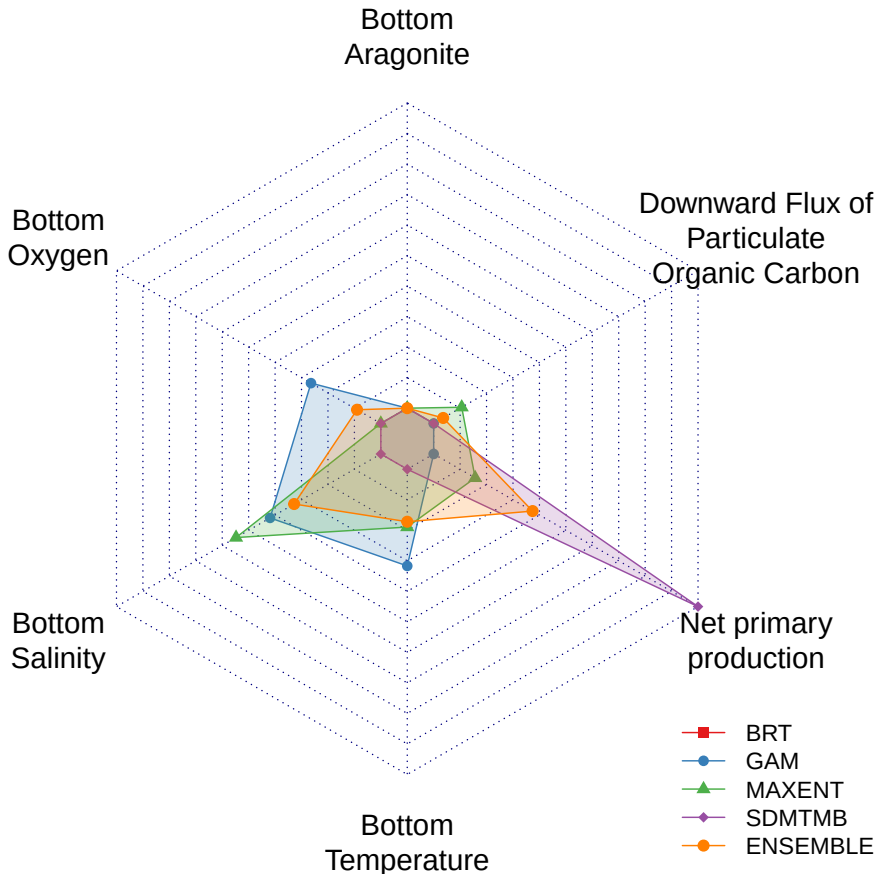


Figure 10: Dynamic Factors driving Atlantic surfclam distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature, bottom salinity, and net primary productivity were the most important to species distribution (Figure 10). Surfclams are known to be sensitive to thermal stress and occupy coastal habitats (Cargnelli et al. 1999). They are generalist planktivores (Cargnelli et al. 1999). When exposure was calculated with only these factors, it remained Moderate on average throughout the species range and spatial patterns remained consistent (Figure 11; Figure 12).

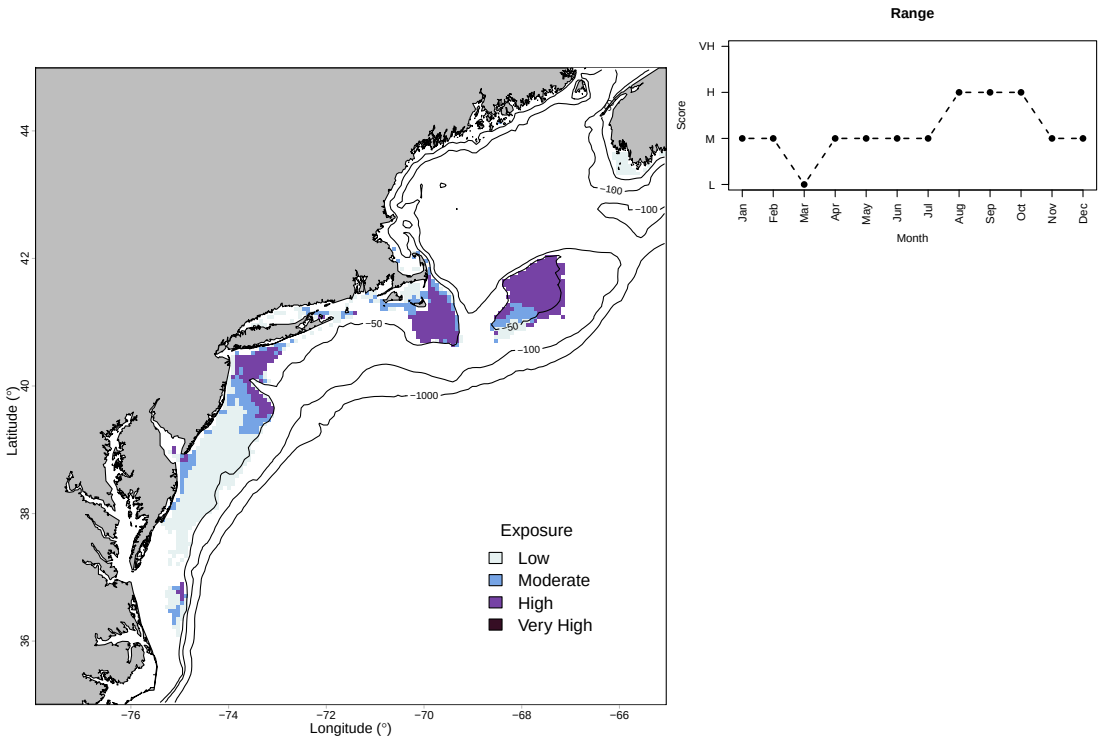


Figure 11: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Atlantic surfclam distributions

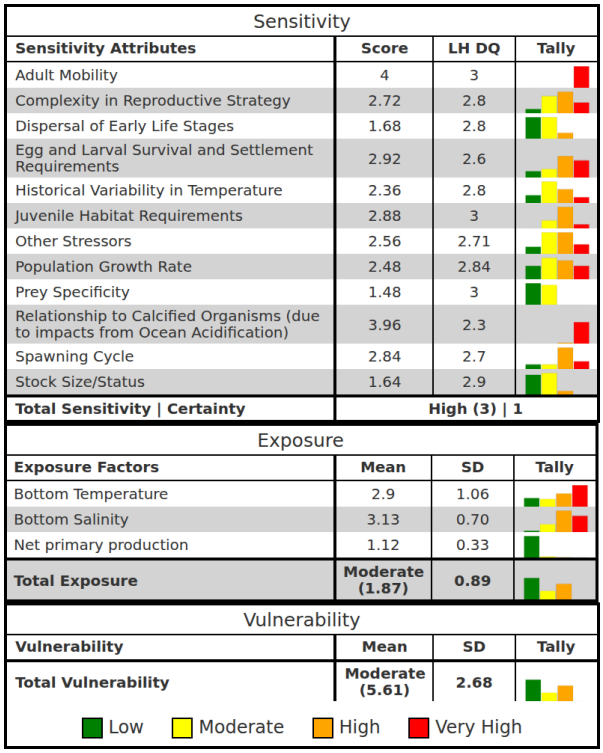


Figure 12: As in Figure 1 for Atlantic surfclam, but with only significant Exposure factors and the resulting Vulnerability Scores.

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